

Exploratory Data Analysis (EDA) Questions

Complete the following tasks using pandas, NumPy, and visualization libraries. Write your own code in the cells provided below. Do not include answers; only solve each question by writing code.

Setup

Import the required Python libraries for data analysis and visualization.

```
In [3]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

Load the Dataset

Load the provided CSV dataset into a pandas DataFrame and display the first five records to verify that it has been loaded correctly.

```
In [4]: df = pd.read_csv(r"C:\Users\sharm\OneDrive\Documents\ai_workforce_displacement_g
```

```
In [5]: df.head()
```

```
Out[5]:
```

	record_id	country	iso3_code	region	income_group	year	quarter	quarter_label
0	1	United States	USA	North America	High Income	2020	1	2020-Q1
1	2	United States	USA	North America	High Income	2020	1	2020-Q1
2	3	United States	USA	North America	High Income	2020	1	2020-Q1
3	4	United States	USA	North America	High Income	2020	1	2020-Q1
4	5	United States	USA	North America	High Income	2020	1	2020-Q1

5 rows × 23 columns



Question 1: Dataset Overview

Perform an initial exploration of the dataset. Write code to determine:

- Number of rows and columns
- Data types of all columns
- Summary statistics for numerical features
- Number of unique values in each column
- Missing values in each column
- Duplicate records in the dataset.

```
In [6]: df.shape
```

```
Out[6]: (20800, 23)
```

```
In [7]: df.dtypes
```

```
Out[7]: record_id          int64
country          object
iso3_code         object
region           object
income_group      object
year             int64
quarter          int64
quarter_label     object
industry_sector   object
sector_automation_risk_score float64
gdp_per_capita_usd int64
ai_adoption_index float64
pct_sector_workforce_displaced float64
pct_sector_workforce_new_roles_created float64
net_workforce_change_pct float64
ai_cited_layoff_announcements int64
ai_skill_wage_premium_pct float64
pct_workforce_female float64
pct_displaced_roles_female float64
reskilling_programs_count int64
govt_ai_policy_score_1_to_10 float64
ai_tool_adoption_pct float64
data_source_notes object
dtype: object
```

```
In [8]: df.describe()
```

Out[8]:

	record_id	year	quarter	sector_automation_risk_score	gdp_per
count	20800.000000	20800.000000	20800.000000	20800.000000	20
mean	10400.500000	2022.769231	2.423077	0.537881	21
std	6004.587135	1.887406	1.115411	0.190345	22
min	1.000000	2020.000000	1.000000	0.151000	
25%	5200.750000	2021.000000	1.000000	0.379000	3
50%	10400.500000	2023.000000	2.000000	0.580000	10
75%	15600.250000	2024.000000	3.000000	0.698000	40
max	20800.000000	2026.000000	4.000000	0.855000	99

In [9]: `df.nunique()`

```
Out[9]: record_id          20800
country             80
iso3_code           80
region             12
income_group        4
year                7
quarter            4
quarter_label       26
industry_sector     10
sector_automation_risk_score  679
gdp_per_capita_usd  1981
ai_adoption_index   775
pct_sector_workforce_displaced  2068
pct_sector_workforce_new_roles_created  1680
net_workforce_change_pct    837
ai_cited_layoff_announcements  519
ai_skill_wage_premium_pct   490
pct_workforce_female       10
pct_displaced_roles_female  453
reskilling_programs_count   91
govt_ai_policy_score_1_to_10  78
ai_tool_adoption_pct    799
data_source_notes         1
dtype: int64
```

In [10]: `df.duplicated().sum()`Out[10]: `np.int64(0)`In [11]: `df.isnull().sum()`

```
Out[11]: record_id          0
country          0
iso3_code        0
region           0
income_group     0
year             0
quarter          0
quarter_label    0
industry_sector  0
sector_automation_risk_score  0
gdp_per_capita_usd  0
ai_adoption_index  0
pct_sector_workforce_displaced  0
pct_sector_workforce_new_roles_created  0
net_workforce_change_pct  0
ai_cited_layoff_announcements  0
ai_skill_wage_premium_pct  0
pct_workforce_female  0
pct_displaced_roles_female  0
reskilling_programs_count  0
govt_ai_policy_score_1_to_10  0
ai_tool_adoption_pct  0
data_source_notes  0
dtype: int64
```

Question 2

Count the number of unique countries.

```
In [12]: df['country'].nunique()
```

```
Out[12]: 80
```

Question 3

Display all unique industry sectors.

```
In [13]: df['industry_sector'].unique()
```

```
Out[13]: array(['Technology & Software', 'Finance & Banking',
                'Healthcare & Life Sciences', 'Manufacturing & Industry',
                'Retail & E-Commerce', 'Education & Research',
                'Transportation & Logistics', 'Media & Communications',
                'Administrative & Clerical', 'Energy & Utilities'], dtype=object)
```

Question 4

Find the average AI Adoption Index.

```
In [14]: df['ai_adoption_index'].mean()
```

```
Out[14]: np.float64(0.6921477884615385)
```

Question 5

Which country has the highest GDP per capita?

```
In [18]: df.groupby('country')['gdp_per_capita_usd'].mean().sort_values(ascending=False).
```

```
Out[18]:
```

	country	gdp_per_capita_usd
0	Switzerland	90747.153846

Question 6

Find the top 10 countries with the highest AI Adoption Index.

```
In [19]: df.groupby('country')['ai_adoption_index'].mean().sort_values(ascending=False).r
```

```
Out[19]:
```

	country	ai_adoption_index
0	Singapore	0.905346
1	Israel	0.885842
2	United States	0.885408
3	Sweden	0.880223
4	Denmark	0.872588
5	Switzerland	0.872473
6	South Korea	0.868415
7	Finland	0.867404
8	China	0.860900
9	Norway	0.859692

Question 7

Find the average AI Adoption Index by region.

```
In [20]: df.groupby('region')['ai_adoption_index'].mean().reset_index()
```

Out[20]:

	region	ai_adoption_index
0	Africa	0.537141
1	Central Asia	0.609708
2	East Asia	0.848202
3	Europe	0.796663
4	Europe/Asia	0.698483
5	Latin America	0.641559
6	Middle East	0.735563
7	Middle East/Africa	0.574108
8	North America	0.863188
9	Oceania	0.827458
10	South Asia	0.570106
11	Southeast Asia	0.662900

Question 8

Find the average workforce displacement by industry.

```
In [21]: df.groupby('industry_sector')['pct_sector_workforce_displaced'].mean().reset_index
```

Out[21]:

	industry_sector	pct_sector_workforce_displaced
0	Administrative & Clerical	0.092016
1	Education & Research	0.025171
2	Energy & Utilities	0.041974
3	Finance & Banking	0.060348
4	Healthcare & Life Sciences	0.020882
5	Manufacturing & Industry	0.079422
6	Media & Communications	0.054123
7	Retail & E-Commerce	0.073076
8	Technology & Software	0.045995
9	Transportation & Logistics	0.068542

Question 9

Which industry has the highest automation risk?

```
In [22]: df.groupby('industry_sector')['sector_automation_risk_score'].mean().sort_values
```

```
Out[22]: industry_sector
Administrative & Clerical    0.789742
Name: sector_automation_risk_score, dtype: float64
```

Question 10

Find countries with AI Adoption Index greater than 0.8.

```
In [23]: df1 = df.groupby('country')['ai_adoption_index'].mean().reset_index()
```

```
In [24]: df1[df1['ai_adoption_index'] > 0.8]
```

```
Out[24]:
```

	country	ai_adoption_index
2	Australia	0.827315
3	Austria	0.817785
5	Belgium	0.822019
10	Canada	0.840969
12	China	0.860900
16	Denmark	0.872588
20	Finland	0.867404
21	France	0.817712
22	Germany	0.834996
26	Hong Kong	0.855296
32	Israel	0.885842
35	Japan	0.809450
44	Netherlands	0.854769
45	New Zealand	0.827600
47	Norway	0.859692
60	Singapore	0.905346
62	South Korea	0.868415
65	Sweden	0.880223
66	Switzerland	0.872473
67	Taiwan	0.846946
72	UAE	0.834019
74	United Kingdom	0.848312
75	United States	0.885408

Question 11

Display only records from 2025./

In [25]: `df[df['year']==2025]`

Out[25]:

	record_id	country	iso3_code	region	income_group	year	quarter	quarter_label
200	201	United States	USA	North America	High Income	2025	1	2025
201	202	United States	USA	North America	High Income	2025	1	2025
202	203	United States	USA	North America	High Income	2025	1	2025
203	204	United States	USA	North America	High Income	2025	1	2025
204	205	United States	USA	North America	High Income	2025	1	2025
...	...	...	...	...	...	...	...	...
20775	20776	Zambia	ZMB	Africa	Lower Middle Income	2025	4	2025
20776	20777	Zambia	ZMB	Africa	Lower Middle Income	2025	4	2025
20777	20778	Zambia	ZMB	Africa	Lower Middle Income	2025	4	2025
20778	20779	Zambia	ZMB	Africa	Lower Middle Income	2025	4	2025
20779	20780	Zambia	ZMB	Africa	Lower Middle Income	2025	4	2025

3200 rows × 23 columns



Question 12

Find the average wage premium for each income group.

In [26]: `df.groupby('income_group')['ai_skill_wage_premium_pct'].mean().reset_index()`

Out[26]:

	income_group	ai_skill_wage_premium_pct
0	High Income	0.394390
1	Low Income	0.310399
2	Lower Middle Income	0.330390
3	Upper Middle Income	0.356316

Question 13

Count observations for each region.

In [27]: `df['region'].value_counts()`

```
Out[27]: region
Europe                5200
Latin America         3380
Africa                3380
Southeast Asia       2080
Middle East           1820
East Asia             1300
South Asia            1300
North America         520
Oceania               520
Europe/Asia           520
Central Asia          520
Middle East/Africa    260
Name: count, dtype: int64
```

Question 14

Find the top five countries with the highest workforce displacement.

In [28]: `df.groupby('country')['pct_sector_workforce_displaced'].mean().sort_values(ascen`

```
Out[28]:
```

	country	pct_sector_workforce_displaced
0	Singapore	0.106867
1	Denmark	0.104123
2	United States	0.103910
3	Finland	0.103638
4	Sweden	0.103595

Question 15

Calculate the correlation matrix.

In [29]: `df.corr(numeric_only=True)`

Out[29]:

	record_id	year	quarter	sector_ai
record_id	1.000000	1.235380e-02	3.732868e-04	
year	0.012354	1.000000e+00	-1.180546e-01	
quarter	0.000373	-1.180546e-01	1.000000e+00	
sector_automation_risk_score	-0.000042	-7.469622e-04	-2.721199e-04	
gdp_per_capita_usd	-0.444218	4.806326e-02	3.429363e-04	
ai_adoption_index	-0.404061	5.542376e-01	2.853618e-02	
pct_sector_workforce_displaced	-0.291737	4.502595e-01	8.420588e-03	
pct_sector_workforce_new_roles_created	-0.325253	3.360655e-01	5.721231e-03	
net_workforce_change_pct	0.076561	-5.142227e-01	-1.096248e-02	
ai_cited_layoff_announcements	-0.077123	5.143983e-01	1.245326e-02	
ai_skill_wage_premium_pct	-0.165218	8.778019e-01	-5.913468e-02	
pct_workforce_female	0.000023	9.760530e-17	-9.829869e-18	
pct_displaced_roles_female	0.000331	-6.459176e-04	6.253342e-04	
reskilling_programs_count	-0.379482	2.635605e-01	-3.143848e-02	
govt_ai_policy_score_1_to_10	-0.376267	2.175426e-01	-2.539006e-02	
ai_tool_adoption_pct	-0.206153	2.882533e-01	1.498766e-02	

Question 16

Find the average AI adoption for each year.

```
In [30]: df.groupby('year')['ai_adoption_index'].mean()
```

```
Out[30]: year
2020    0.542641
2021    0.610068
2022    0.667465
2023    0.717097
2024    0.758661
2025    0.794465
2026    0.817129
Name: ai_adoption_index, dtype: float64
```

Question 17

Find the average net workforce change by industry.

```
In [31]: df.groupby('industry_sector')['net_workforce_change_pct'].mean()
```

```
Out[31]: industry_sector
Administrative & Clerical    -0.033699
Education & Research        -0.009323
Energy & Utilities          -0.015178
Finance & Banking           -0.022037
Healthcare & Life Sciences  -0.007478
Manufacturing & Industry    -0.029116
Media & Communications      -0.019889
Retail & E-Commerce         -0.026680
Technology & Software       -0.016672
Transportation & Logistics  -0.025166
Name: net_workforce_change_pct, dtype: float64
```

Question 18

Which region has the highest average government AI policy score?

```
In [32]: df.groupby('region')['govt_ai_policy_score_1_to_10'].mean().sort_values(ascending=True)
```

```
Out[32]: region
North America    7.014615
Name: govt_ai_policy_score_1_to_10, dtype: float64
```

Question 19

Create a new column called AI Readiness Score.

AI Readiness = AI Adoption + Policy Score

```
In [33]: df['ai_readiness_score'] = df['ai_adoption_index'] + df['govt_ai_policy_score_1_to_10']
```

```
Out[33]: 0      7.230
1      7.317
2      7.616
3      7.618
4      7.028
...
20795   4.203
20796   4.911
20797   5.009
20798   4.902
20799   4.818
Name: ai_readiness_score, Length: 20800, dtype: float64
```

Question 20

Display countries where workforce displacement exceeds new roles created.

```
In [34]: df2 = df.groupby('country')['pct_sector_workforce_displaced'].mean().reset_index()
```

```
In [35]: df3 = df.groupby('country')['pct_sector_workforce_new_roles_created'].mean().res
```

```
In [36]: df4 = pd.merge(df2,df3,how='left',on = 'country')
```

```
In [37]: df4[df4['pct_sector_workforce_displaced']>df4['pct_sector_workforce_new_roles_cr
```

```
Out[37]:
```

	country	pct_sector_workforce_displaced	pct_sector_workforce_new_roles_created
0	Algeria	0.030793	0.010816
1	Argentina	0.041730	0.022870
2	Australia	0.098747	0.074518
3	Austria	0.097350	0.072977
4	Bangladesh	0.028423	0.010081
...	...	...	...
75	United States	0.103910	0.081212
76	Uruguay	0.051993	0.039654
77	Uzbekistan	0.029628	0.010620
78	Vietnam	0.034400	0.011930
79	Zambia	0.025905	0.008975

80 rows × 3 columns

Question 21

Find the average reskilling programs count by income group.

```
In [41]: df.groupby('income_group')['reskilling_programs_count'].mean().sort_values(ascen
```

```
Out[41]:
```

	income_group	reskilling_programs_count
0	High Income	59.348511
1	Upper Middle Income	28.823427
2	Lower Middle Income	11.481090
3	Low Income	3.457692

Question 22

Find the year with the highest average AI Adoption.

```
In [42]: df.groupby('year')['ai_adoption_index'].mean().sort_values(ascending=False).head
```

Out[42]:

	year	ai_adoption_index
0	2026	0.817129

Question 23

Display the top five records with the highest AI wage premium.

In [43]: `df.nlargest(5, 'ai_skill_wage_premium_pct')`

Out[43]:

	record_id	country	iso3_code	region	income_group	year	quarter	quarter_label
5700	5701	Denmark	DNK	Europe	High Income	2026	1	2026
14039	14040	UAE	ARE	Middle East	High Income	2026	2	2026
16362	16363	Hong Kong	HKG	East Asia	High Income	2026	1	2026
5195	5196	Sweden	SWE	Europe	High Income	2026	2	2026
12218	12219	Portugal	PRT	Europe	High Income	2026	2	2026

5 rows × 24 columns

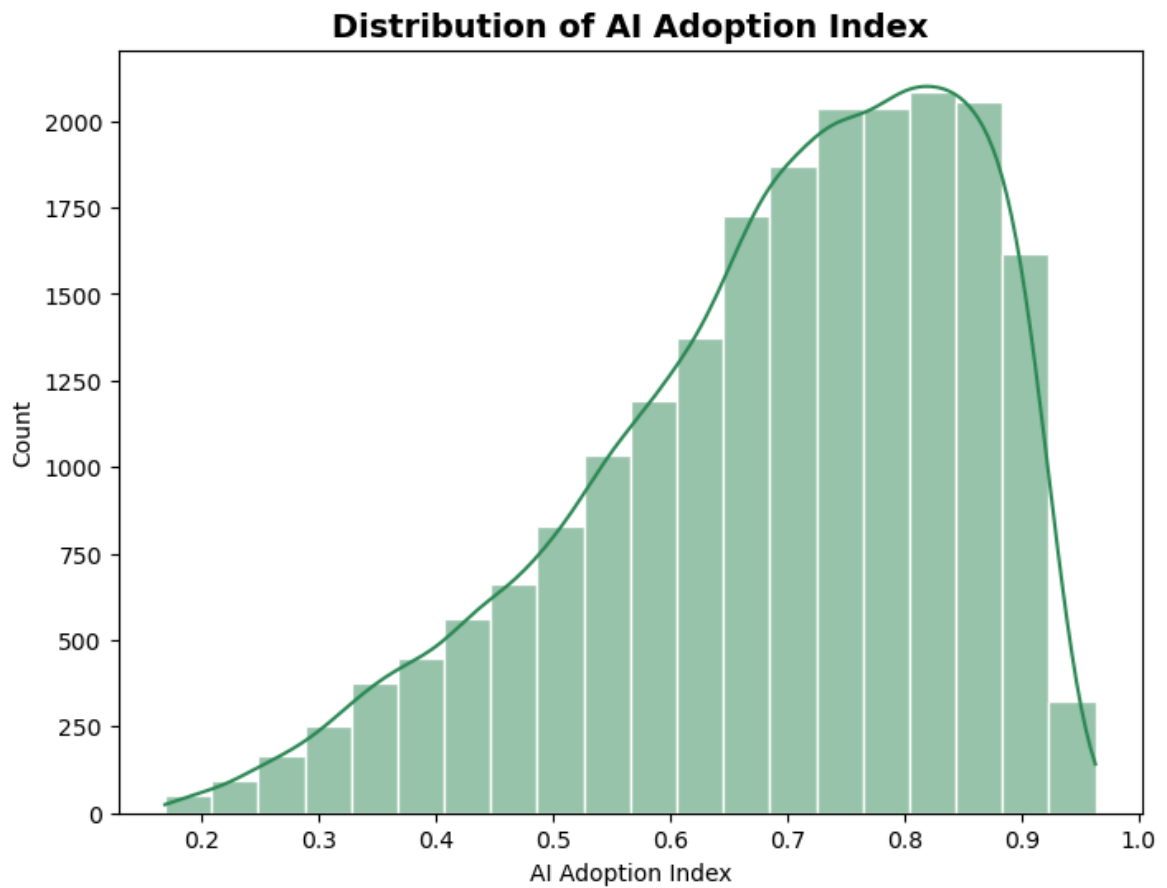


Question 24

Distribution of AI Adoption Index

In [51]: `plt.figure(figsize=(8,6))
sns.histplot(df['ai_adoption_index'],color='seagreen',bins=20,edgecolor='white',
plt.title("Distribution of AI Adoption Index",fontsize=14,fontweight='bold')
plt.xlabel("AI Adoption Index")
plt.ylabel("Count")`

Out[51]: Text(0, 0.5, 'Count')

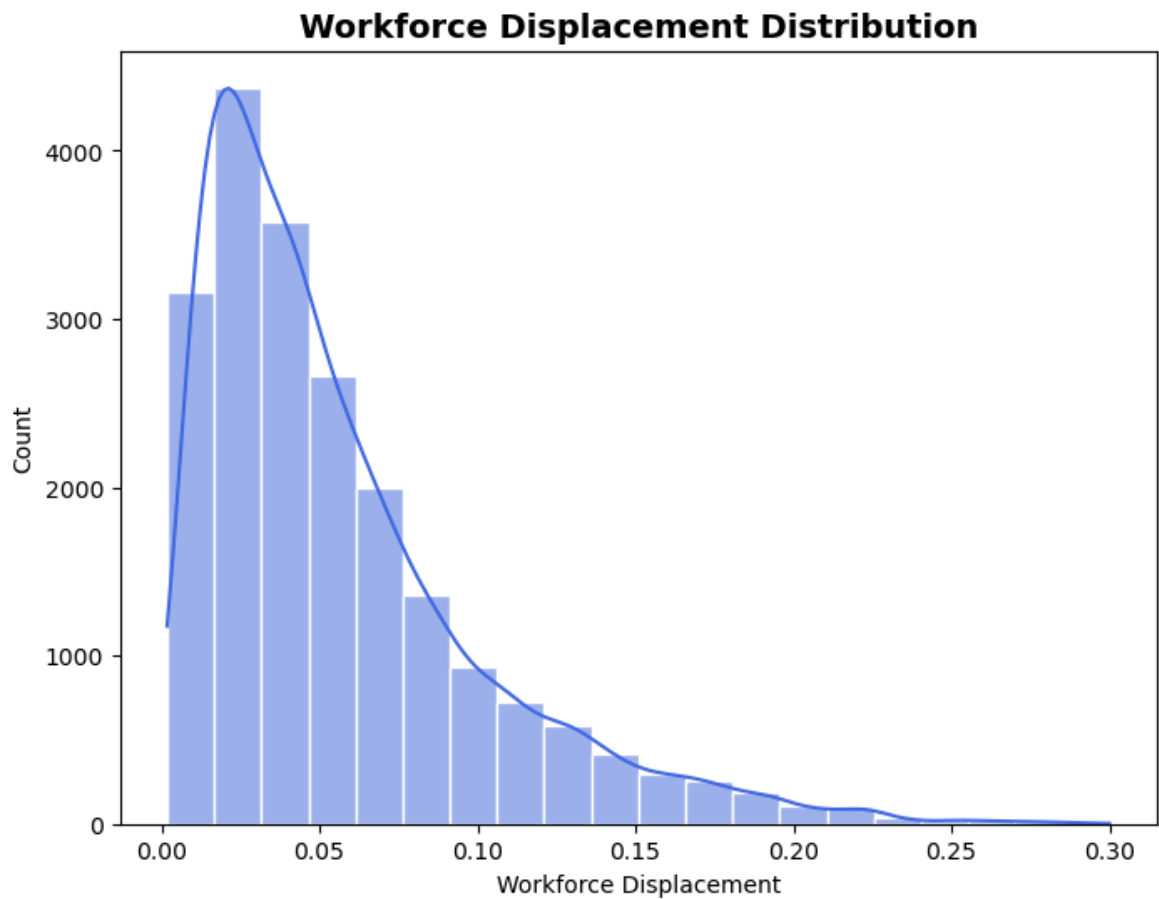


Question 25

Workforce Displacement Distribution

```
In [52]: plt.figure(figsize=(8,6))
sns.histplot(df['pct_sector_workforce_displaced'],color='royalblue',bins=20,edge
plt.title("Workforce Displacement Distribution",fontsize=14,fontweight='bold')
plt.xlabel("Workforce Displacement")
plt.ylabel("Count")
```

Out[52]: Text(0, 0.5, 'Count')

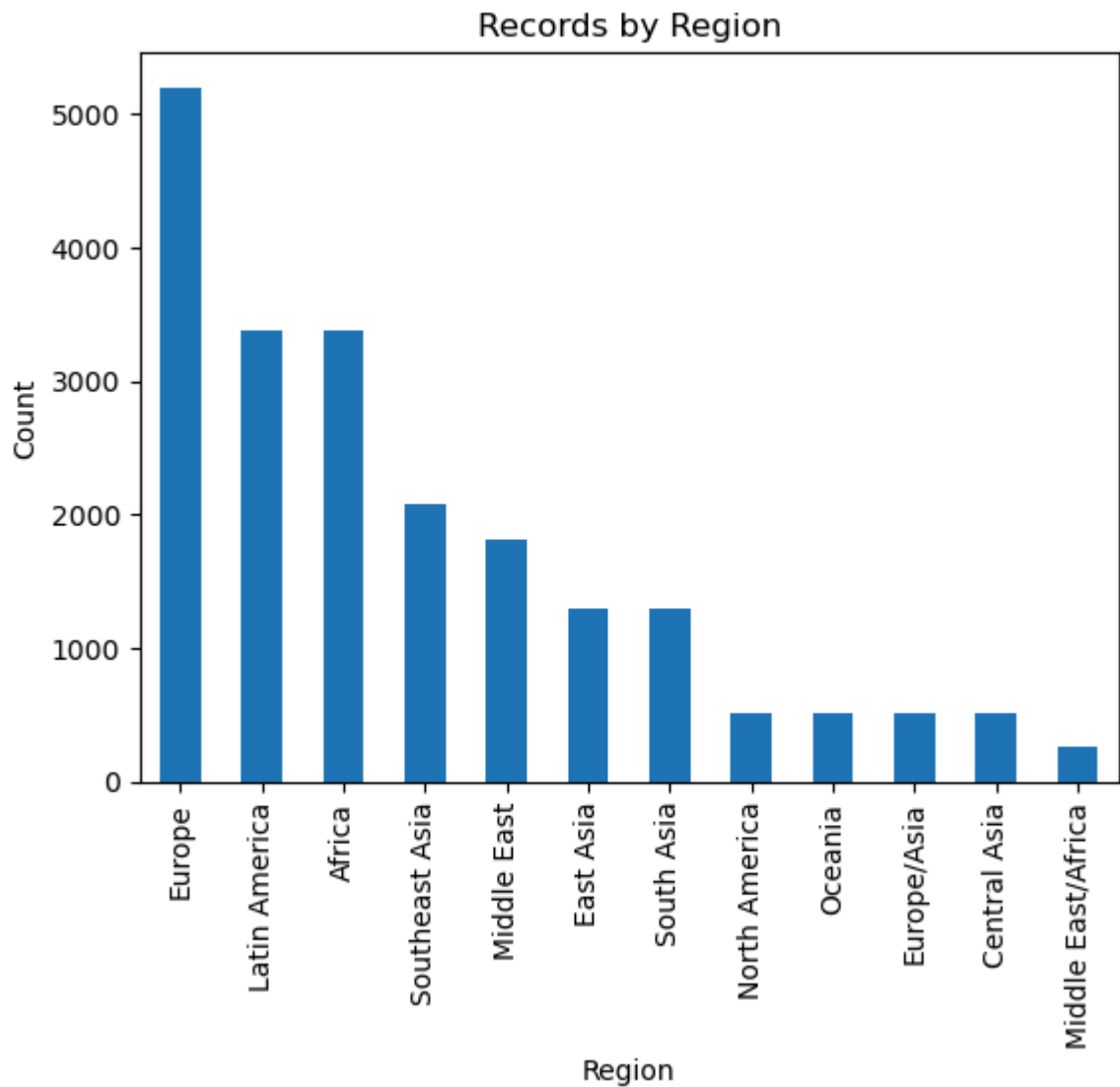


Question 26

Records by Region

```
In [54]: df['region'].value_counts().plot(kind='bar')
plt.title("Records by Region")
plt.xlabel("Region")
plt.ylabel("Count")
```

```
Out[54]: Text(0, 0.5, 'Count')
```

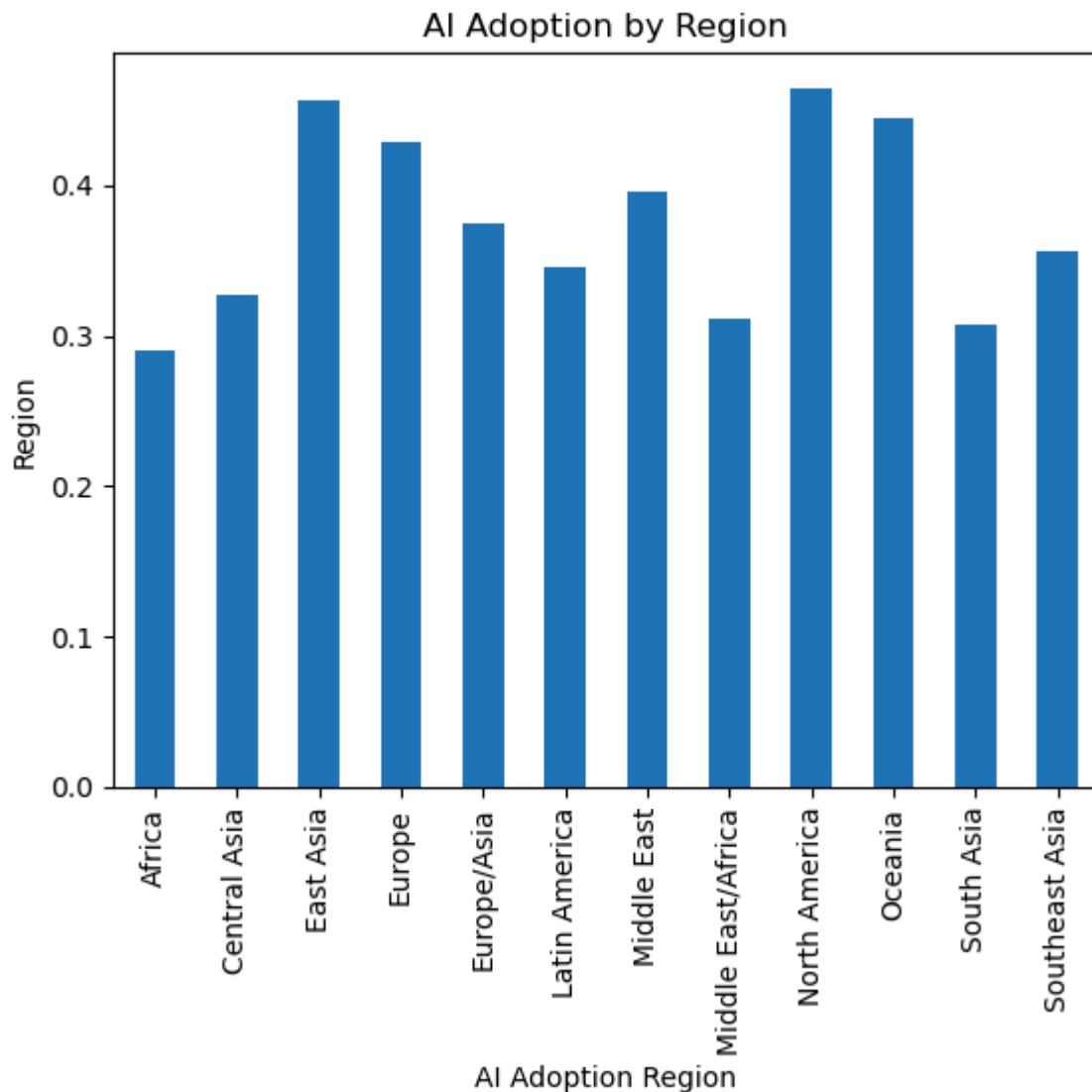


Question 27

AI Adoption by Region

```
In [56]: df.groupby('region')['ai_tool_adoption_pct'].mean().plot(kind='bar')
plt.title("AI Adoption by Region")
plt.xlabel("AI Adoption Region")
plt.ylabel("Region")
```

```
Out[56]: Text(0, 0.5, 'Region')
```

Question 28

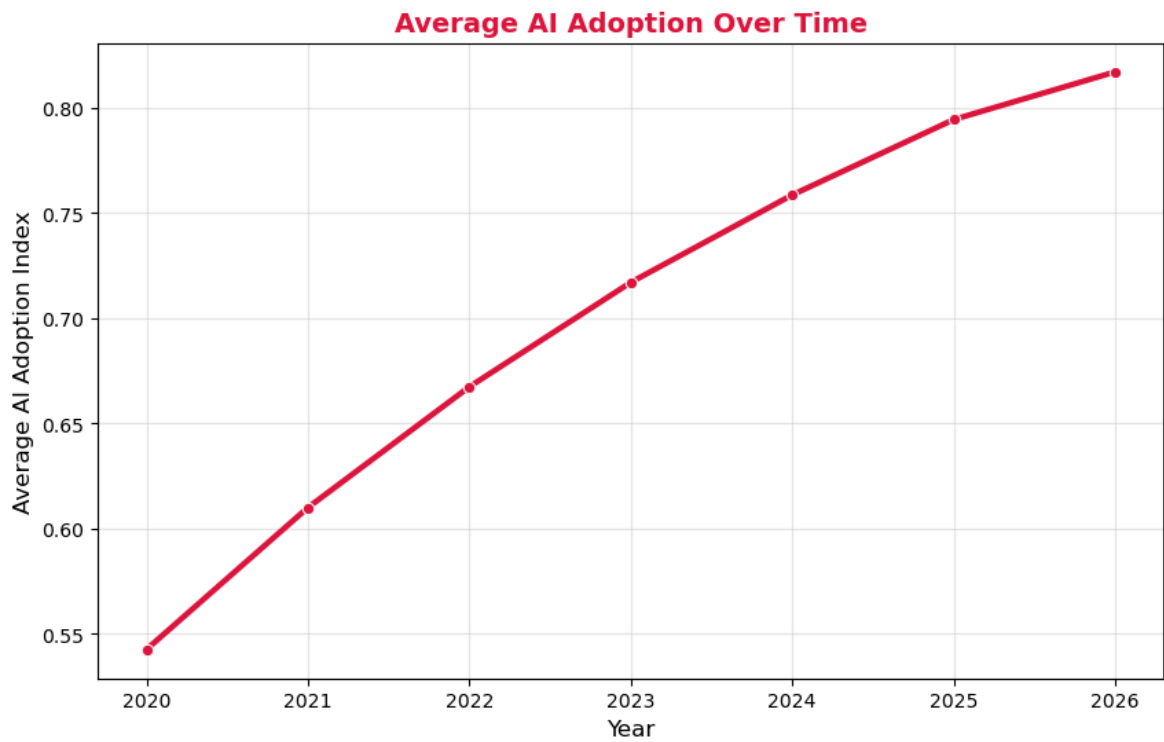
AI Adoption Over Time

```
In [57]: plt.figure(figsize=(10,6))

ai_year = df.groupby('year')['ai_adoption_index'].mean().reset_index()

sns.lineplot(
    data=ai_year,
    x='year',
    y='ai_adoption_index',
    marker='o',
    linewidth=3,
    color='crimson'
)

plt.title('Average AI Adoption Over Time', fontsize=14, fontweight='bold', color='crimson')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Average AI Adoption Index', fontsize=12)
plt.grid(alpha=0.3)
plt.show()
```



Question 29

Workforce Displacement Over Time

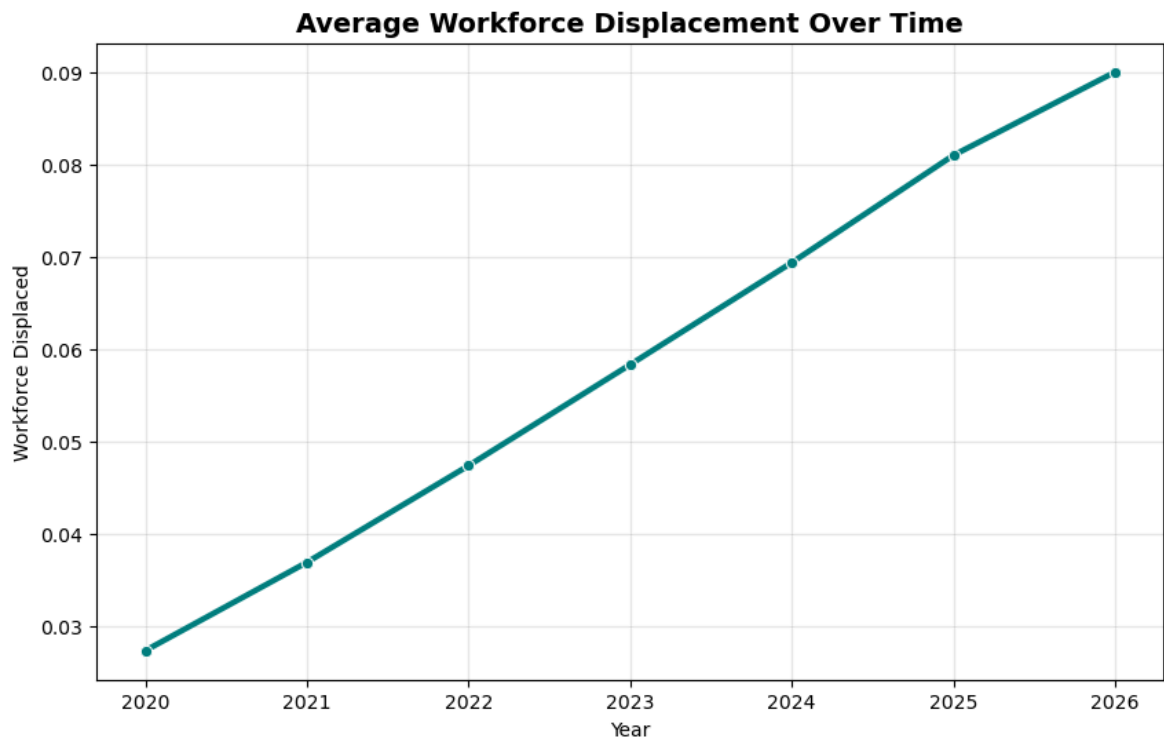
```
In [58]: plt.figure(figsize=(10, 6))

workforce = df.groupby('year')['pct_sector_workforce_displaced'].mean().reset_in

sns.lineplot(
    data=workforce,
    x='year',
    y='pct_sector_workforce_displaced',
    marker='o',
    linewidth=3,
    color='teal'
)

plt.title('Average Workforce Displacement Over Time', fontsize=14, fontweight='b')
plt.xlabel('Year')
plt.ylabel('Workforce Displaced')

plt.grid(alpha=0.3)
plt.show()
```



Question 30

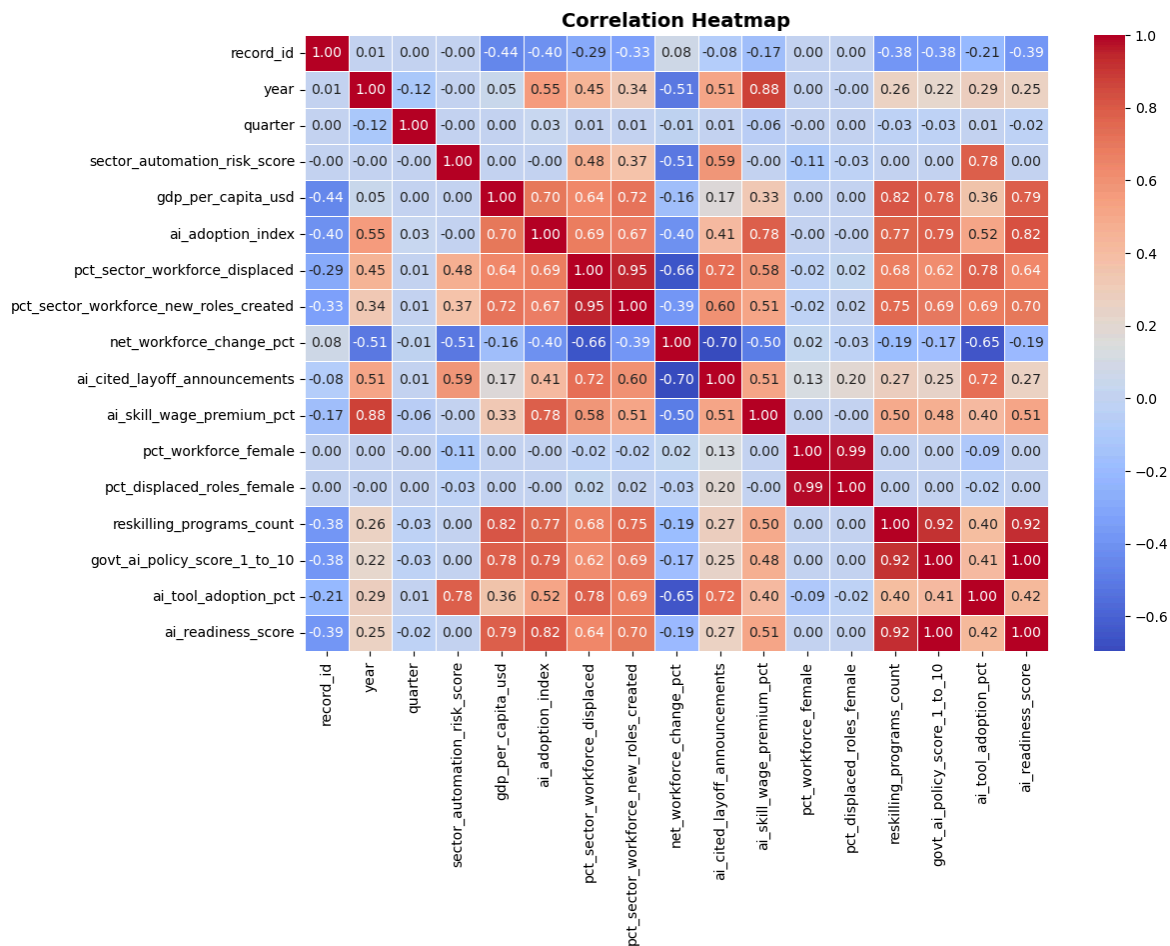
Correlation Heatmap

```
In [59]: plt.figure(figsize=(12, 8))

corr = df.corr(numeric_only=True)

sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm',
    linewidths=0.5,
    fmt='.2f'
)

plt.title('Correlation Heatmap', fontsize=14, fontweight='bold')
plt.show()
```



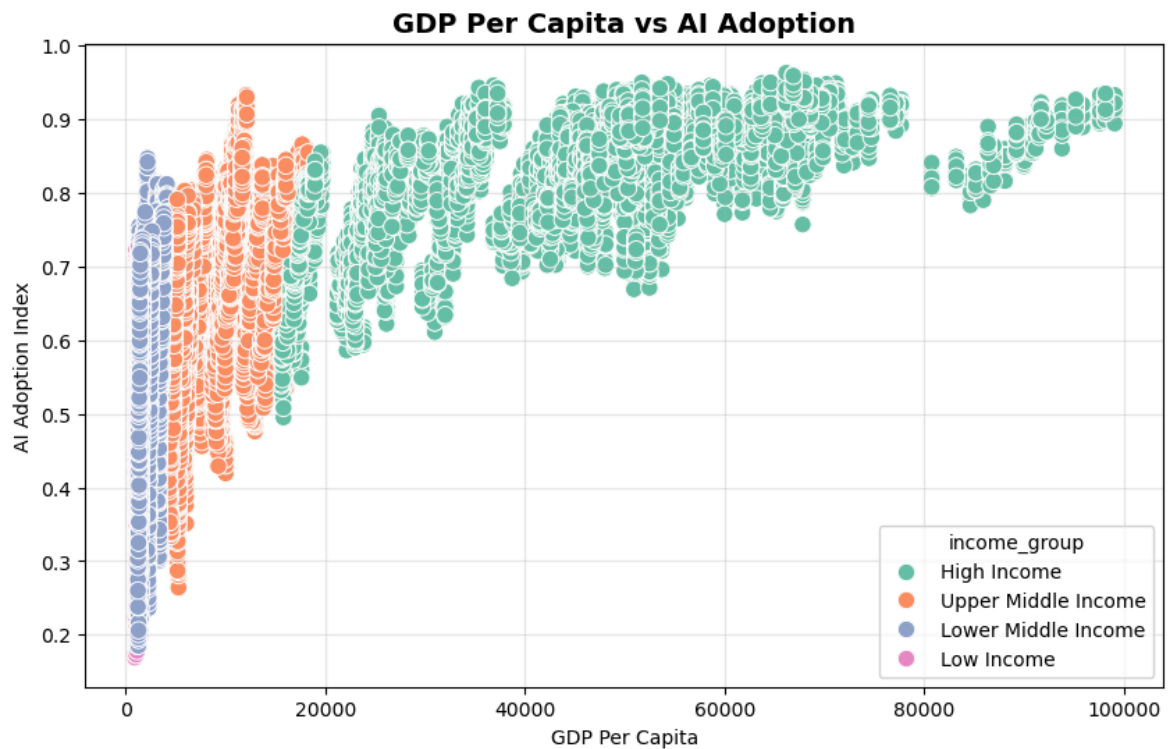
Question 31

GDP vs AI Adoption

```
In [60]: plt.figure(figsize=(10,6))

sns.scatterplot(
    data=df,
    x='gdp_per_capita_usd',
    y='ai_adoption_index',
    hue='income_group',
    palette='Set2',
    s=80
)

plt.title('GDP Per Capita vs AI Adoption', fontsize=14, fontweight='bold')
plt.xlabel('GDP Per Capita')
plt.ylabel('AI Adoption Index')
plt.legend(title='income_group')
plt.grid(alpha=0.3)
plt.show()
```



Question 32

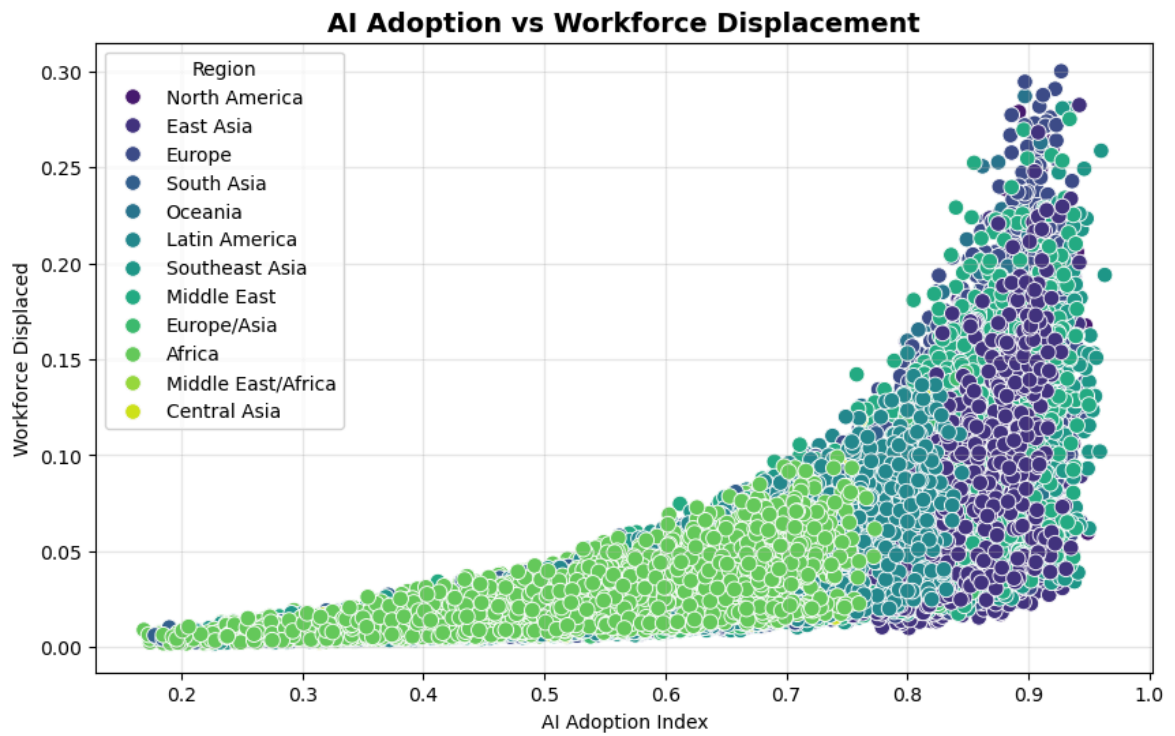
AI Adoption vs Workforce Displacement - Scatter Plot

```
In [61]: plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df,
    x='ai_adoption_index',
    y='pct_sector_workforce_displaced',
    hue='region',
    palette='viridis',
    s=70
)

plt.title('AI Adoption vs Workforce Displacement', fontsize=14, fontweight='bold')
plt.xlabel('AI Adoption Index')
plt.ylabel('Workforce Displaced')
plt.legend(title='Region')

plt.grid(alpha=0.3)
plt.show()
```



Question 33

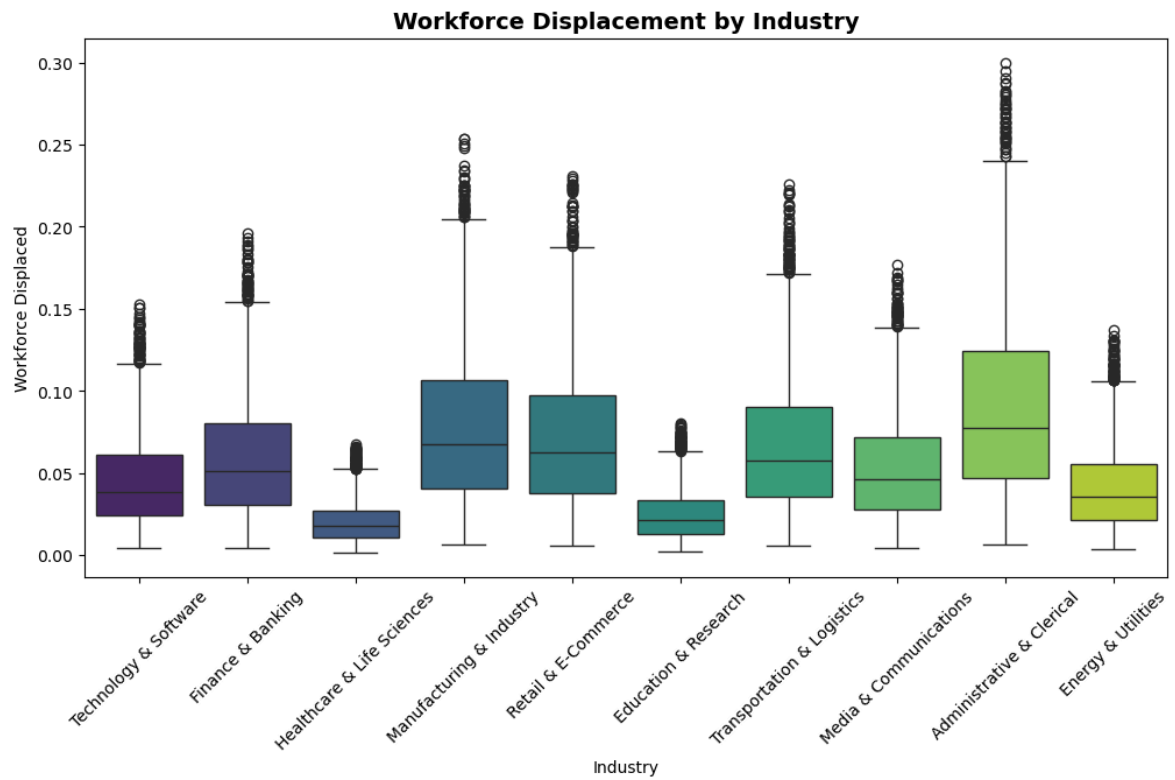
Workforce Displacement by Industry - boxplot

```
In [64]: plt.figure(figsize=(12,6))

sns.boxplot(
    data=df,
    x='industry_sector',
    y='pct_sector_workforce_displaced',
    hue='industry_sector',
    palette='viridis',
    legend=False
)

plt.title('Workforce Displacement by Industry', fontsize=14, fontweight='bold')
plt.xlabel('Industry')
plt.ylabel('Workforce Displaced')
plt.xticks(rotation=45)

plt.show()
```



Question 34

Automation Risk by Industry (Bar Chart)

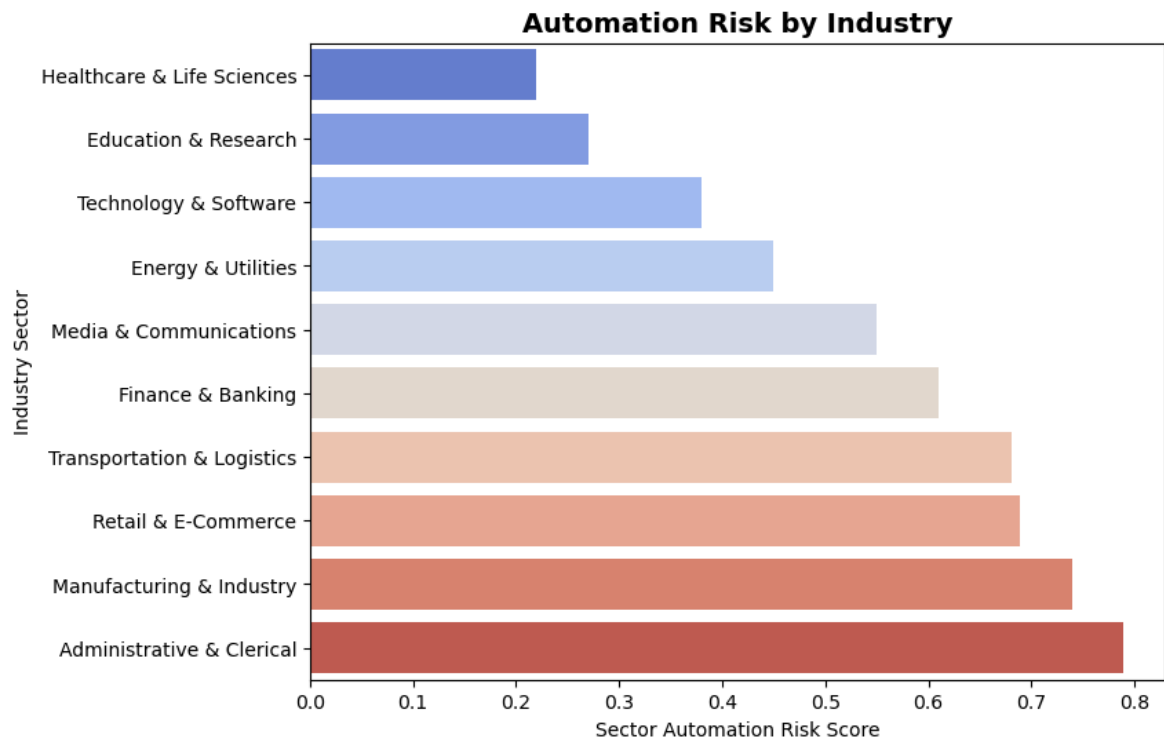
```
In [65]: plt.figure(figsize=(8,6))

risk = df.groupby('industry_sector')['sector_automation_risk_score'].mean().sort

sns.barplot(
    data=risk.reset_index(),
    x='sector_automation_risk_score',
    y='industry_sector',
    hue='industry_sector',
    palette='coolwarm',
    legend=False
)

plt.title('Automation Risk by Industry', fontsize=14, fontweight='bold')
plt.xlabel('Sector Automation Risk Score')
plt.ylabel('Industry Sector')

plt.show()
```



Question 35

Top 10 Countries by AI Adoption

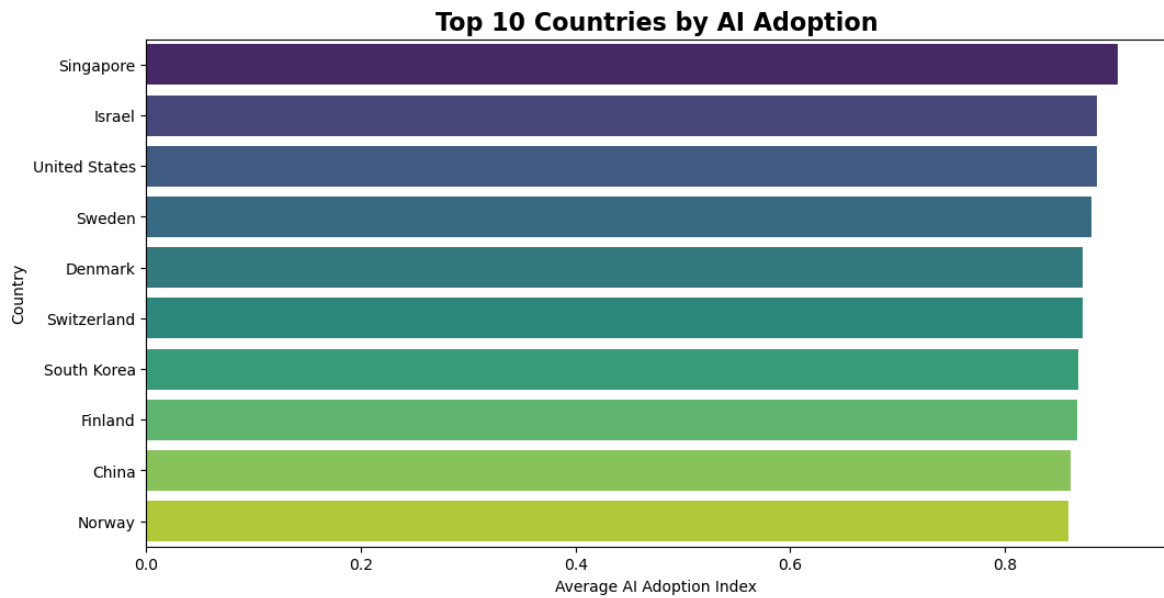
```
In [66]: plt.figure(figsize=(12,6))

top10 = df.groupby('country')['ai_adoption_index'].mean().sort_values(ascending=

sns.barplot(
    data=top10.reset_index(),
    x='ai_adoption_index',
    y='country',
    hue='country',
    palette='viridis',
    legend=False
)

plt.title('Top 10 Countries by AI Adoption', fontsize=16, fontweight='bold')
plt.xlabel('Average AI Adoption Index')
plt.ylabel('Country')

plt.show()
```

Question 36

AI Wage Premium by Income Group

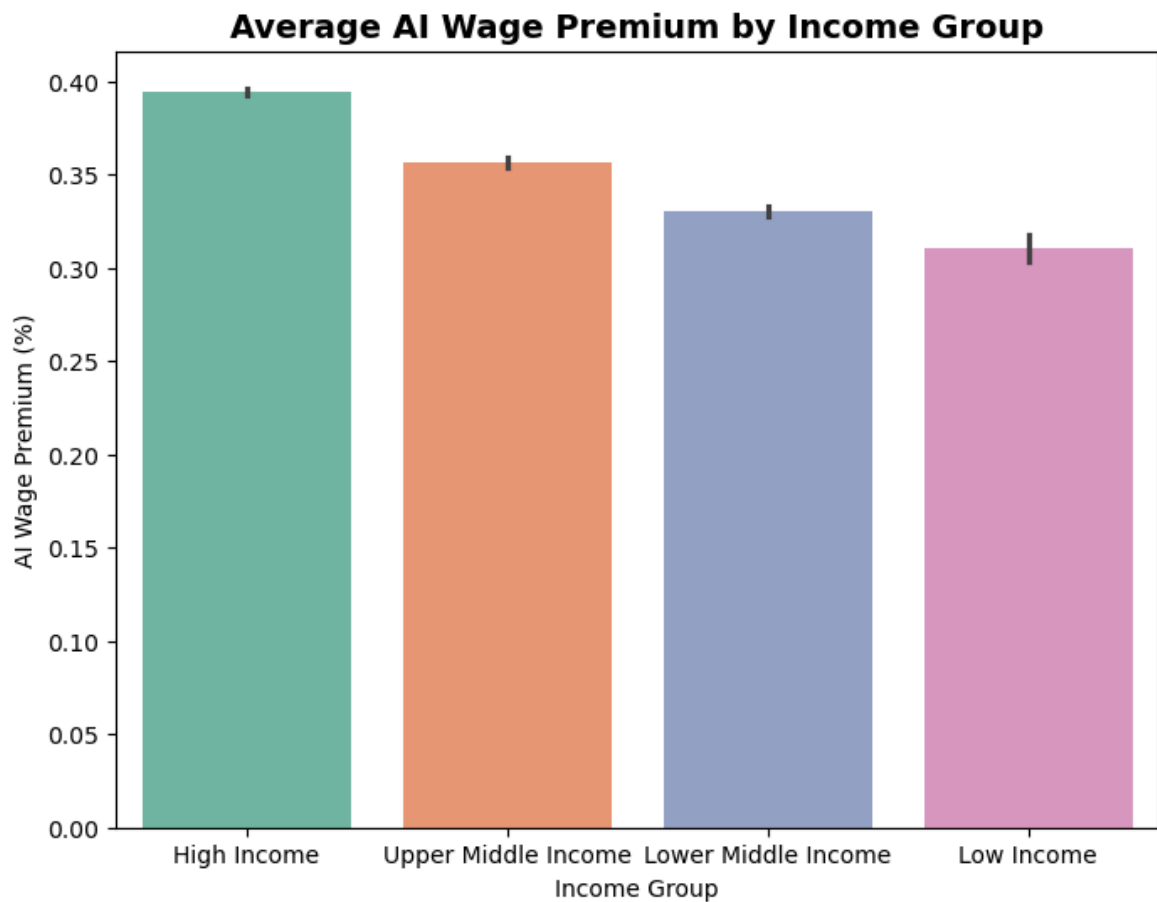
```
In [67]: plt.figure(figsize=(8,6))

sns.barplot(
    data=df,
    x='income_group',
    y='ai_skill_wage_premium_pct',
    hue='income_group',
    palette='Set2',
    legend=False
)

plt.title('Average AI Wage Premium by Income Group',
          fontsize=14,
          fontweight='bold')

plt.xlabel('Income Group')
plt.ylabel('AI Wage Premium (%)')

plt.show()
```



Question 37

Female Workforce vs Female Displacement

```
In [68]: plt.figure(figsize=(6,4))

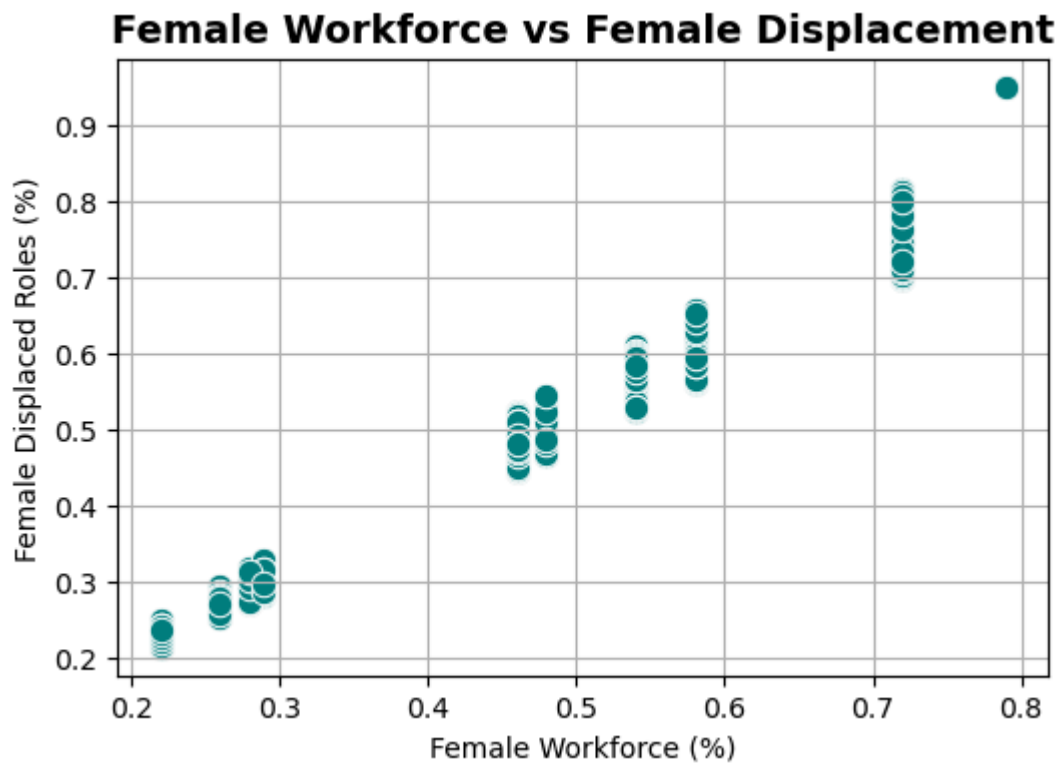
sns.scatterplot(
    data=df,
    x='pct_workforce_female',
    y='pct_displaced_roles_female',
    color='teal',
    s=80
)

plt.grid(True)

plt.title(
    "Female Workforce vs Female Displacement",
    fontsize=14,
    fontweight='bold'
)

plt.xlabel('Female Workforce (%)')
plt.ylabel('Female Displaced Roles (%)')

plt.show()
```



Question 38

AI Tool Adoption by Income Group

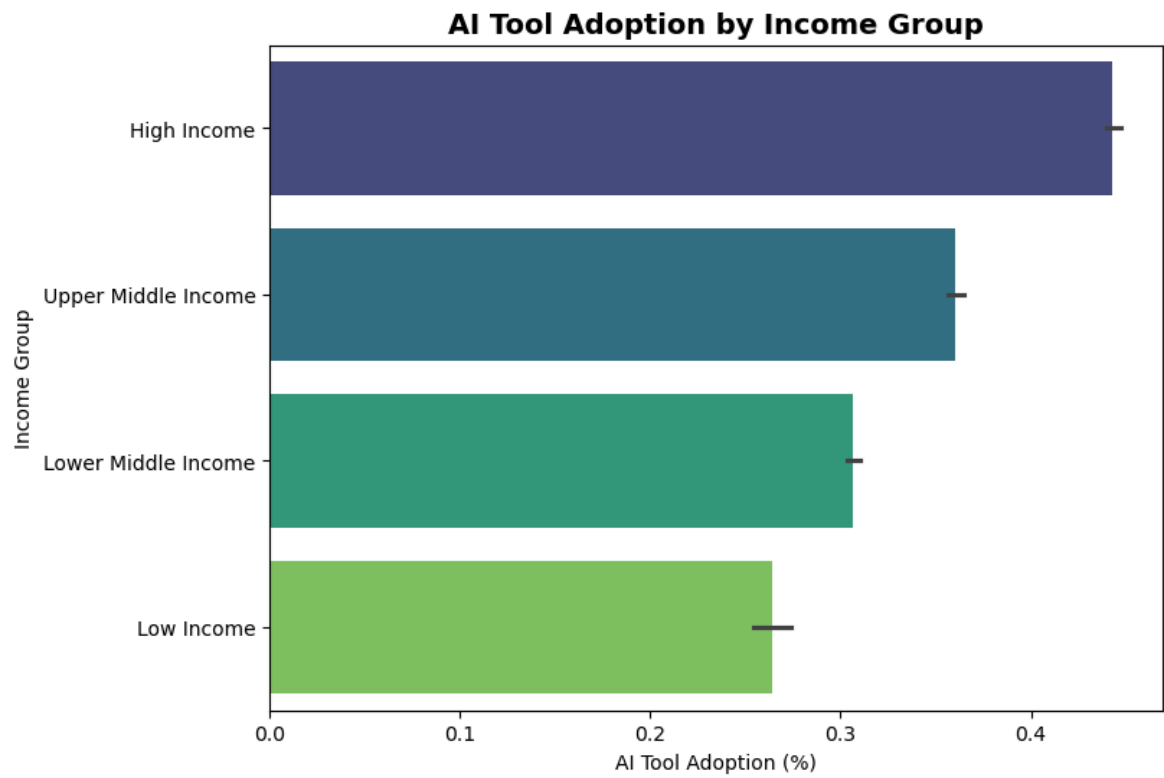
```
In [69]: plt.figure(figsize=(8,6))

sns.barplot(
    data=df,
    x='ai_tool_adoption_pct',
    y='income_group',
    hue='income_group',
    palette='viridis',
    legend=False
)

plt.title('AI Tool Adoption by Income Group',
          fontsize=14,
          fontweight='bold')

plt.xlabel('AI Tool Adoption (%)')
plt.ylabel('Income Group')

plt.show()
```



In []: